Numerical Exercise #3

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1 Question #1 Numerical formulation

Relationship between $\Phi_i,\ \Phi_{i+1},\ \Phi_{i-1}$ for group g at any point within the material:

$$\left(-\frac{D_g}{\Delta x^2}\right)\phi_g[i-1]+\left(\frac{2D_g}{\Delta x^2}+\Sigma_{r,g}\right)\phi_g[i]+\left(-\frac{D_g}{\Delta x^2}\right)\phi_g[i+1]=S_g[i]$$

with $\sum_{r,g}$:

$$\Sigma_{r,1}=\Sigma_{a,1}+\Sigma_{s,1\to 2}$$
$$\Sigma_{r,2}=\Sigma_{a,2}$$

Expression of the source term for the fast and thermal energy group:

$$S_1[i]=\nu\Sigma_{f,1}\,\phi_1[i]+\nu\Sigma_{f,2}\,\phi_2[i]$$
$$S_2[i]=\Sigma_{s,1\to 2}\,\phi_1[i]$$

2 Question #2 Analytical solution for the bare system

$$k_{eff}=1.08440 \tag{1}$$

3 Question #3 Numerical solution of the bare homogeneous reactor

Numerical results

keff (scientific format with 5 decimal points):

$$k_{\text{eff}}=1.08677\times 10^{0}$$

Fast buckling (scientific format with 3 decimal points):

$$B_1^2=9.141\times 10^{-4}\ \text{cm}^{-2}$$

Thermal buckling (scientific format with 3 decimal points):

$$B_2^2=9.285\times 10^{-4}\ \text{cm}^{-2}$$

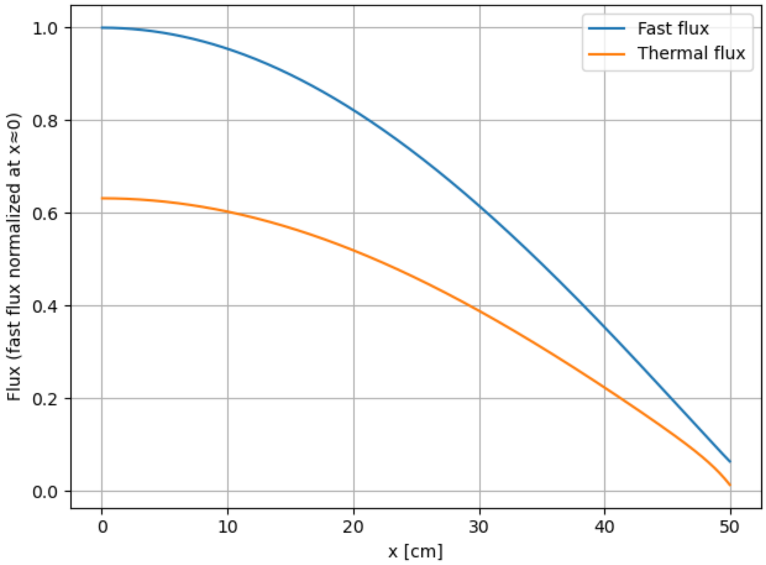

Figure 1: Normalized fast flux and scaled to the same factor thermal flux, (bare core

4 Question #4

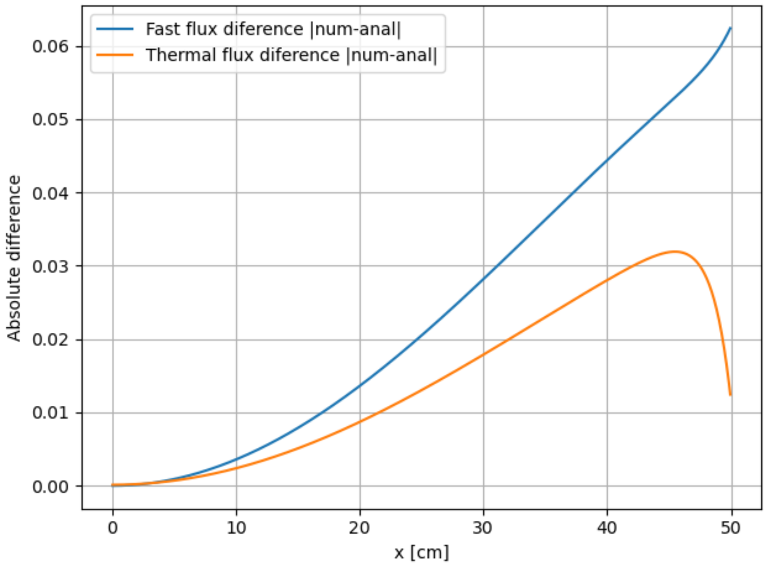

Figure 2: Difference between numerical and analytical solutions for the fast and thermal fluxes in the bare reactor for a mesh size of 0.1 cm.

5 Question #5 Numerical solution of the reflected reactor

Numerical results

keff (scientific format with 5 decimal points):

$$k_{\text{eff}} = 1.09198 \times 10^{0}$$

Fast net current (scientific format with 5 decimal points) at the core/reflector interface:

$$J_1(x = a) = 3.95033 \times 10^{-2}\ \text{cm}^{-2}\text{s}^{-1}$$

Thermal net current (scientific format with 5 decimal points) at the core/reflector interface:

$$J_2(x = a) = -5.86376 \times 10^{-3}\ \text{cm}^{-2}\text{s}^{-1}$$

First harmonic eigenvalue (scientific format with 5 decimal points):

$$k_2 = 1.02112 \times 10^{0}$$

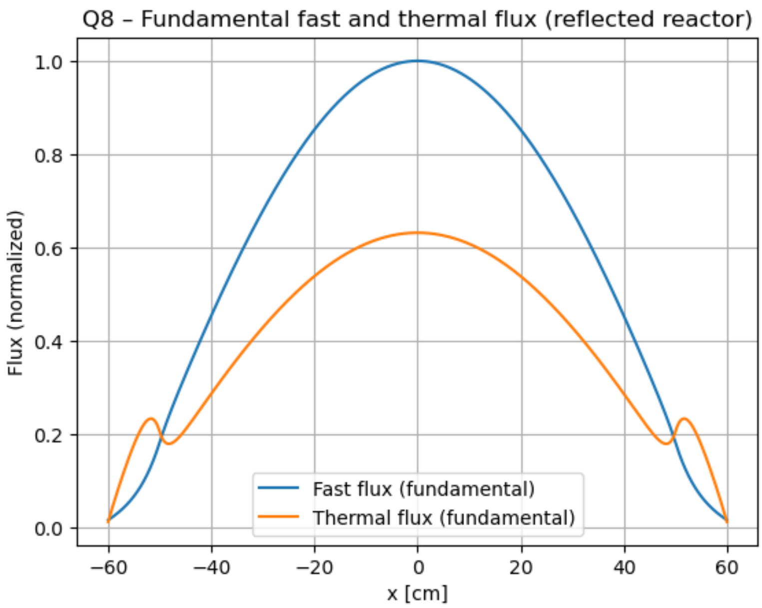

Figure 3: Fast and thermal fundamental fluxes in the reflected reactor for a mesh size of 0.1 cm.

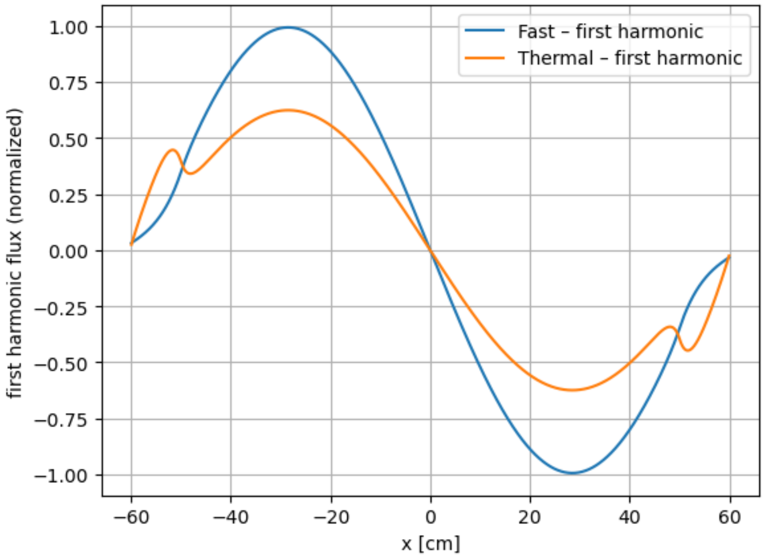

Figure 4: Fast and thermal first harmonic of the fluxes in the reflected reactor for a mesh size of 0.1 cm.